

Software Requirement Specifications

Version: 1.9

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<i>Supervisor</i>	<i>Mr. Basit Ali</i>
<i>Co Supervisor</i>	<i>-</i>
<i>Project Team</i>	<i>Alizah Basit (22K-5155)</i> <i>Amna Mansoor (22K-5159)</i> <i>Hafsa Salman (22K-5161)</i>
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1. Introduction

1.1. Purpose of Document

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements for the product titled “Intelligent System for Pakistan Sign Language (PSL) Learning and Translation”. This SRS specifies the features, behavior, and constraints of the system to ensure a common understanding among stakeholders, developers, and end users.

*The scope of this document covers the entire PSL mobile application, which consists of four core modules: **Dictionary**, **Translation**, **Learning**, and **Assessment**. Each module contributes to the overall goal of providing an accessible, interactive, and reliable platform for PSL translation and education. Future enhancements, such as multi-language support and advanced community features, are outside the scope of this initial release and will be addressed in subsequent versions.*

1.2. Intended Audience

This Software Requirements Specification (SRS) is intended for the following stakeholders:

- **Project Sponsors and Management:** *To understand the project objectives, scope, and resource requirements.*
- **Developers:** *To obtain detailed functional and non-functional requirements necessary for design and implementation.*
- **Tester:** *To use the defined requirements as the basis for preparing test cases and validating system functionality.*
- **UI/UX Designer:** *To align interface designs with the requirements for usability and accessibility.*
- **Documentation Writers:** *To prepare user manuals, help guides, and training materials.*

End Users (Deaf community representatives, educators, and learners): *To validate that the requirements address real-world needs and usability expectations.*

1.3 Abbreviations

- **3D-CNN:** *Three-Dimensional Convolutional Neural Network*
- **ASL:** *American Sign Language*
- **BLEU:** *Bilingual Evaluation Understudy (machine translation score)*
- **BSL:** *British Sign Language*
- **CNN:** *Convolutional Neural Network*
- **CSL:** *Chinese Sign Language*
- **CVPR:** *IEEE Conference on Computer Vision and Pattern Recognition*
- **DGS:** *German Sign Language*
- **DL:** *Deep Learning*
- **GFSLT-VLP:** *Gloss-Free Sign Language Translation – Visual-Language Pretraining*
- **HMM:** *Hidden Markov Model*
- **HOG:** *Histogram of Oriented Gradients*
- **ICASSP:** *IEEE International Conference on Acoustics, Speech, and Signal Processing*
- **LSTM:** *Long Short-Term Memory*

- *MCL: Multi-Cue Learning*
- *ML: Machine Learning*
- *MTL: Multi-Task Learning*
- *PSL: Pakistan Sign Language*
- *RGB: Red, Green, Blue*
- *SIFT: Scale-Invariant Feature Transform*
- *SLR: Sign Language Recognition*
- *SLT: Sign Language Translation*
- *SRS: Software Requirements Specifications*
- *SVM: Support Vector Machine*
- *TPAMI: IEEE Transactions on Pattern Analysis and Machine Intelligence*
- *VLP: Visual–Language Pretraining*

1.4 Document Convention

- *Font Family = Arial*
- *Font Size = 12 for headings, 10 for the rest of the content*

2. Overall System Description

2.1. Project Background

Communication between hearing and non-hearing individuals remains a significant barrier in Pakistan, primarily due to the limited availability of tools that support Pakistan Sign Language (PSL). Unlike widely supported sign languages such as ASL or BSL, PSL suffers from a lack of digital resources, structured learning platforms, and real-time translation systems. As a result, individuals from the Deaf community often struggle to communicate effectively in educational institutions, workplaces, and public spaces, leading to social exclusion and reduced opportunities.

Existing applications and systems that support PSL are either incomplete, non-interactive, or restricted to simple word lists without real-time gesture recognition. Moreover, there is no unified mobile platform in Pakistan that combines PSL translation, dictionary access, structured learning modules, and assessment tools. This gap highlights the need for a technologically advanced, AI-powered solution that can enhance accessibility, increase awareness, and promote inclusivity.

To address this challenge, the proposed project leverages modern technologies such as computer vision, deep learning, and mobile development. By integrating an encoder–decoder–based translation system, a structured dictionary, and an engaging learning experience, the application aims to bridge the communication gap and provide a reliable, user-friendly, and accessible platform for both the Deaf community and hearing individuals.

Problem Statement:

- *Limited availability of digital tools that support Pakistan Sign Language (PSL).*
- *No real-time PSL translation system exists for everyday communication.*
- *Current PSL resources lack structure, interactivity, and accessibility.*
- *Lack of platforms for learning, practicing, and assessing PSL skills.*
- *Communication barriers contribute to social exclusion for individuals with hearing impairments.*

2.2. Literature review

Sign languages exhibit unique grammatical structures and rely heavily on non-manual markers, making SLT fundamentally different from spoken language translation [3]. PSL also has its own grammar and vocabulary, distinct from English or Urdu [4].

Early SLR research used handcrafted features, which were effective for isolated signs but inadequate for continuous recognition [5]. Later, deep learning models such as CNNs, LSTMs, and 3D-CNNs significantly advanced sign recognition by learning complex spatial–temporal representations [6]. However, these models struggled with long-term dependencies, prompting a shift to transformer-based systems.

Transformers became the state-of-the-art approach for SLT, starting with the work of Camgöz et al. [7],

who developed the first end-to-end translation model for DGS using attention mechanisms. Subsequent research improved translation accuracy through multi-cue fusion integrating hand, face, and body features [9], hierarchical segmentation [10], and pose-aware modeling [11]. Multi-task learning further boosted performance by training recognition, glossing, and translation jointly [8].

Dataset availability is a major factor influencing SLT progress. While datasets such as PHOENIX-2014T [7], CSL-Daily [1], WLASL [12], and How2Sign [13] support ASL, CSL, and DGS research, PSL lacks sentence-level datasets entirely. Existing PSL resources are limited to alphabets or isolated signs, making PSL one of the least developed sign languages in computational research [2].

Recent methods attempt to address low-resource limitations. Sign Back-Translation (SignBT) synthesizes artificial sign-text pairs from monolingual text and significantly improves translation in low-resource settings [1]. Gloss-level augmentation strategies help generalization by expanding or generating synthetic gloss sequences [14, 15]. Gloss-free visual–language models such as GFSLT-VLP eliminate the need for gloss annotations, which is highly beneficial for languages like PSL [16]. Transfer learning from large natural video datasets or multilingual text corpora also provides substantial improvement when sign-specific data is limited [17].

Existing sign language learning systems focus mainly on ASL or ISL using gesture recognition, pose estimation, or hybrid classifier–CNN methods [18, 19, 20]. However, none integrate continuous PSL recognition, PSL–text translation, structured lessons, and assessments into a single application. This highlights both a technical and application-level research gap that the present project intends to address.

2.3. Project Scope

The product, “Intelligent System for PSL Learning and Translation”, is a mobile application designed to bridge the communication gap between hearing and non-hearing individuals in Pakistan. The system combines four key modules—**Dictionary, Translation, Learning, and Assessment**—to deliver both real-time PSL translation and an interactive learning experience.

The primary purpose of the application is to promote accessibility and inclusivity by facilitating effective communication between hearing and non-hearing individuals. Beyond translation, the integrated learning and assessment modules provide structured lessons, quizzes, and progress tracking to encourage long-term adoption and mastery of PSL.

This product aligns with broader organizational and societal goals of improving accessibility, fostering inclusivity, and leveraging technology for social good. By providing a cost-effective and scalable solution, it supports the strategic vision of increasing awareness and adoption of PSL within educational institutions, workplaces, and communities.

Included Functionalities:

- **Dictionary Module** for structured PSL reference
- **Translation Module** for real-time PSL to text conversion
- **Learning Module** for structured tutorials and lessons
- **Assessment Module** for quizzes, tests, and progress tracking

2.4. Not In Scope

The following items are **outside the scope** of this project and will **not** be included in the current version of the system:

- **Support for international sign languages** such as ASL, BSL, or ISL—only Pakistan Sign Language (PSL) will be covered.
- **Full-body gesture recognition** for complex sentences; the system will primarily focus on upper-body and hand-sign recognition.
- **High-fidelity 3D avatar generation** beyond basic animations or prerecorded videos.
- **Web application or desktop version**—the system will be available **only as a mobile application**.
- **Advanced community features**, such as public chatrooms or forums, beyond basic leaderboard or profile interactions.
- **Integration with government institutions**, schools, or official certification bodies.
- **Real-time group communication support** (e.g., group video calls with live PSL translation).
- **Speech-to-speech translation** or multilingual translation beyond English and Urdu.

2.5. Project Objectives

User Functions:

The application provides the following core functions:

- **PSL Dictionary:**
 - Allows users to search and browse PSL words alphabetically or chapter-wise.
 - Provides gesture videos for each word.
- **Translation Module:**
 - Converts PSL gestures (captured via camera) into text or speech.
- **Learning Module:**
 - Offers structured lessons organized into chapters.
 - Displays PSL gesture videos alongside text for guided learning.
- **Assessment Module:**
 - Conducts quizzes, practice exercises, and full-chapter tests.
 - Grades user performance based on accuracy of sign replication or translation.
- **User Management and Engagement:**
 - Provides account creation, login, and profile management.
 - Tracks user progress and engagement with streaks, and rewards

2.6. Stakeholders

- Mobile Application Developer
- Supervisor
- Deaf Community Representatives
- End Users
- UI/UX Designer
- Tester
- FAST NUCES

2.7. Operating Environment

The app will work on **Android phones and tablets** with:

- **Android 9.0 or above**
- **At least 4 GB RAM**
- **A camera and microphone**

It is designed to run smoothly on modern Android devices.

2.8. System Constraints

- The system must use **Firestore and Firebase Storage** for data management and media access.
- AI models must be developed using **TensorFlow/Keras or PyTorch**, with **TFLite** for mobile deployment.
- Real-time translation and data sync require **stable internet connectivity**.
- Performance may be limited by **mobile device hardware**, affecting inference speed and video rendering.
- The application is restricted to **Android devices only** in the current scope.
- All PSL signs and learning content must be **culturally accurate** and approved by the Deaf community.
- User data must follow **secure handling practices** under Firebase Authentication.
- The UI must be **simple, visual, and accessible** for users with varying technical abilities.

2.9. Assumptions & Dependencies

- Users will have access to Android devices running **Android 9.0 or higher** with sufficient hardware resources (≥ 4 GB RAM, camera, and microphone).
- A **stable internet connection** will be available for real-time translation, cloud storage, and updates.
- PSL datasets can be collected, augmented, and used for training AI models without major legal or licensing restrictions.
- Users are expected to have **basic smartphone literacy** to interact with the app.

2.10. Dependencies

- Dependence on third-party frameworks and libraries such as TensorFlow/PyTorch, OpenCV, and MediaPipe for AI and computer vision tasks.
- Dependence on Firebase for data storage, user authentication, and synchronization.
- Dependence on cloud infrastructure for backend services and model deployment.
- Dependence on external APIs for certain translation features.
- Future system upgrades depend on the availability of additional PSL datasets and feedback from the Deaf community.

3. External Interface Requirements

3.1. Hardware Interfaces

The “Intelligent System for Pakistan Sign Language (PSL) Learning and Translation” application will interact with the following hardware components of Android devices:

- **Camera Interface**
 - Used for capturing video input of PSL gestures.
 - Supports both front and rear cameras.
 - Provides real-time frame data for gesture recognition using computer vision models.
- **Touchscreen Interface**
 - Facilitates user interaction with the application (taps, swipes, gestures).
 - Provides input for navigation, text entry, and learning activities.
- **Network Interface (Wi-Fi/Mobile Data)**
 - Enables communication with backend servers and databases.
 - issues.

3.2. Software Interfaces

The “Intelligent System for Pakistan Sign Language (PSL) Learning and Translation” will interact with several software components, frameworks, and services to perform its functions.

- **Operating System:**
 - Android 9.0 (Pie) or higher.
 - Provides core services (camera, storage, networking).
- **Database Interfaces:**
 - **Firebase** for storing user accounts, progress data, dictionary entries, and app content.
 - Data exchanged via secure API calls (JSON format).
- **Backend Frameworks:**
 - **Python** API services.
 - Handles communication between the mobile app and AI models.
- **AI/ML Libraries:**
 - **TensorFlow/Keras, PyTorch** for gesture recognition and translation.
 - **OpenCV, MediaPipe** for real-time hand and pose tracking.
 - **Transformers, NLTK** for natural language processing.
- **Authentication and Security:**
 - **Firebase Authentication** (email/password, possibly social login).
 - Secure communication through HTTPS and token-based authentication (JWT)

The system relies on **RESTful APIs** for communication between components, ensuring modularity and scalability.

3.3. Communications Interfaces

The “Intelligent System for Pakistan Sign Language (PSL) Learning and Translation” requires communication between the mobile application, backend server, and cloud database. The following standards and protocols will be used:

- **Network Communication:**
 - The app will use **Wi-Fi or mobile data (4G/5G)** for connectivity.
 - All communication between client and server will follow **RESTful API standards**.
- **Protocols and Standards:**
 - **HTTP/HTTPS** will be used for all client–server communication.
 - **WebSockets** may be used for real-time interactions (e.g., instant feedback during learning or assessment).
- **Message Formats:**
 - Data exchange will use **JSON** for lightweight and structured communication.
 - Multimedia data (images, video frames) will be transmitted in **compressed formats** (JPEG/MP4).
- **Security and Encryption:**
 - All communications will be secured via **HTTPS/TLS encryption**.
 - **Token-based authentication (JWT)** will be implemented for session security.
- **Data Synchronization:**
 - Offline caching will allow temporary storage of progress/lessons.
 - Data will automatically **sync with Firebase** when the device reconnects to the internet.

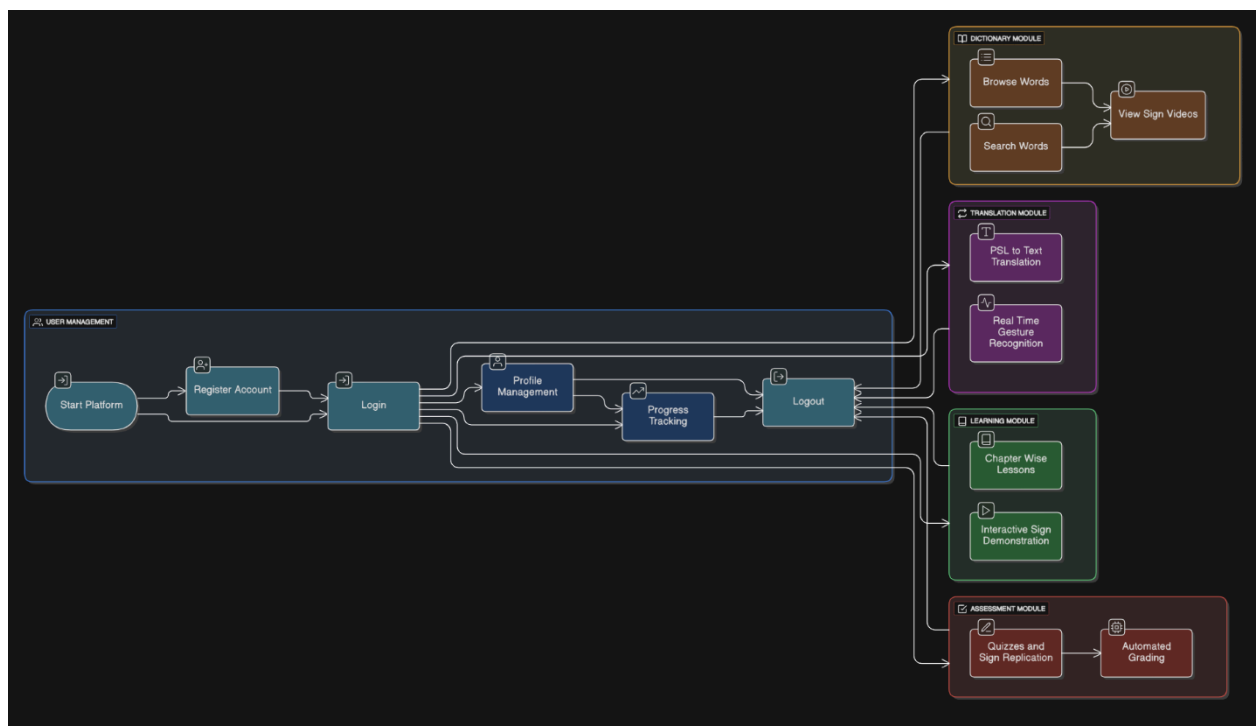
This ensures that the application provides **secure, reliable, and efficient communication** across all modules.

4. Functional Requirements

4.1. Functional Hierarchy

1. User Management
 - 1.1. User Registration
 - 1.2. Login / Logout
 - 1.3. Profile Management
 - 1.4. Progress Tracking
2. Dictionary Module
 - 2.1. Browse PSL words alphabetically or by category
 - 2.2. Search words
 - 2.3. View sign videos
3. Translation Module
 - 3.1. PSL-to-Text Translation
 - 3.2. Real-time gesture recognition using CV
4. Learning Module
 - 4.1. Chapter-wise lessons
 - 4.2. Interactive sign demonstration
5. Assessment Module
 - 5.1. Quizzes and sign replication exercises
 - 5.2. Automated grading using AI model

4.2. Use Case Diagram



UC-User-01	
Use case Id:	UC-User-01
Actors:	User
Feature:	User Registration

Pre-condition:		User must have a phone with proper internet connection
Scenarios		
Step#	Action	Software Reaction
1.	User opens the application	The application will open up and show up the splash screen which lead to the login and sign-up page.
2.	User gives their respective information to sign up.	
Alternate Scenarios: -		
1a: If the information is incomplete, the system prompts the user to add missing information.		
Post Conditions		
Step#	Description	
	User will be signed up successfully.	
Use Case Cross referenced		--

UC-User-02		
Use case Id:	UC-User-02	
Actors:	User	
Feature:	User Logging In	
Pre-condition:	User must have a phone with proper internet connection and must have signed up before	
Scenarios		
Step#	Action	Software Reaction
1.	User opens the application.	The application will open up and show up the splash screen which lead to the login and sign-up page.
2.	User will enter his/her credentials and click log in.	The application will lead the user to the home page.
Alternate Scenarios: -		
1a: If the credentials are incomplete, the system prompts the user to add missing information.		
1b: If the credentials are invalid, the system prompts the user to add valid credentials.		
Post Conditions		
Step#	Description	
	User will be logged in successfully.	

Use Case Cross referenced	--

UC-User-03		
Use case Id:	UC-User-03	
Actors:	User	
Feature:	Dictionary Module	
Pre-condition:	User must have a phone with proper internet connection and a account logged in.	
Scenarios		
Step#	Action	Software Reaction
1.	User opens the application	The application will lead the user to the home page.
2.	User will click the dictionary module button.	The application will lead the user to the dictionary module with words grouped in chapter in alphabetical order.
3.	User can look for the certain word by searching.	The searched word will pop up.
Alternate Scenarios: -		
1a: If the word doesn't exist, the application will say "No word found".		
Post Conditions		
Step#	Description	
	User will successfully go to the dictionary module and can search for the word.	
Use Case Cross referenced	--	

UC-User-04		
Use case Id:	UC-User-04	
Actors:	User	
Feature:	Translation Module	
Pre-condition:	User must have a phone with proper internet connection, and an account logged in.	
Scenarios		
Step#	Action	Software Reaction

1.	User opens the application	The application will lead the user to the home page.
2.	User will click the translation module button.	The application will lead the user to the translation module with camera focused on the user.
3.	User will sign a word.	The application will translate the sign to the word.
Alternate Scenarios: -		
1a: If the word was not trained, the application will say “No word found”.		
Post Conditions		
Step#	Description	
	User will successfully get the translation of the signed word.	
Use Case Cross referenced		--

UC-User-05		
Use case Id:	UC-User-05	
Actors:	User	
Feature:	Learning Module	
Pre-condition:	User must have a phone with proper internet connection and an account logged in.	
Scenarios		
Step#	Action	Software Reaction
1.	User opens the application	The application will lead the user to the home page.
2.	On the home page, user click on the respective level.	The application will lead the user to the learning module's particular chapter where user will learn new sign words.
Alternate Scenarios: -		
Post Conditions		
Step#	Description	
	User will successfully go to the learning module and learn how to sign.	
Use Case Cross referenced		--

UC-User-06

Use case Id:		UC-User-06
Actors: User		
Feature:		Assessment Module
Pre-condition:		User must have a phone with proper internet connection and an account logged in.
Scenarios		
Step#	Action	Software Reaction
1.	User opens the application	The application will lead the user to the home page.
2.	User will click the learning module button.	The application will lead the user to the learning module's particular chapter where user will learn new sign words.
3.	After completion of chapter, an assessment will be taken.	The user will complete the assessment in order for the software to mark it.
Alternate Scenarios: -		
Post Conditions		
Step#	Description	
	User will successfully complete the assessment and will make a progress in learning sign language.	
Use Case Cross referenced		UC-User-05

5. Non-functional Requirements

5.1. Performance Requirements

The "Intelligent System for Pakistan Sign Language (PSL) Learning and Translation" must perform efficiently under real-world usage conditions to ensure reliability and user satisfaction. The following performance requirements are defined:

- **Real-Time Translation:**
 - The system shall process PSL gestures from the camera and return text/speech output within **2–3 seconds**.
- **Scalability:**
 - The backend shall support at least **1,000 concurrent users** without significant performance degradation.
 - Cloud infrastructure must allow horizontal scaling to handle increased demand.
- **Dictionary Lookup:**
 - Word searches shall retrieve and display results in less than **1 second** under normal network conditions.
- **Learning and Assessment Modules:**
 - Lesson content (examples) must load within **2 seconds**.
 - Quizzes and feedback shall be evaluated and displayed within **3 seconds** of user submission.
- **Resource Utilization:**

- The app shall run smoothly on Android devices, ensuring efficient memory and battery usage.
 - AI model computations will be optimized to minimize device overheating and excessive power consumption.
- **System Availability:**

The application shall maintain **99% uptime** for backend services to ensure uninterrupted access for users.

5.2. Safety Requirements

The “Intelligent System for Pakistan Sign Language (PSL) Learning and Translation” must ensure that its use does not result in harm, loss, or misuse of user data and that it complies with basic digital safety standards. The following safety requirements apply:

- **Data Protection:**
 - All user data (profiles, progress, personal details) must be stored securely to prevent unauthorized access or loss.
 - Sensitive data such as passwords must be encrypted both in transit and at rest.
- **User Safeguards:**
 - The system shall prevent unauthorized access by enforcing secure login and authentication protocols.
 - In case of failed login attempts, account lockout or captcha verification must be enforced to prevent misuse.
- **System Integrity:**
 - The application shall prevent data corruption by ensuring that partially completed uploads or interrupted processes are either retried or rolled back safely.
 - System crashes or errors must not result in loss of user progress or stored lessons.
- **Regulatory Compliance:**
 - The application must comply with relevant data privacy and digital safety policies (e.g., GDPR principles for data handling).
 - Any third-party service integrated (e.g., Firebase, APIs) must also meet standard security and safety certifications.
- **User Awareness:**
 - Clear warnings or guidance must be displayed when the device lacks a stable internet connection, preventing incomplete or misleading translations.
 - The system must not provide inaccurate feedback without notifying the user of uncertainty (e.g., low-confidence gesture recognition).

5.3. Security Requirements

The “Intelligent System for Pakistan Sign Language (PSL) Learning and Translation” must ensure secure access, protect user data, and maintain privacy at all times. The following security requirements apply:

- **Authentication and Authorization:**
 - Users must log in using secure credentials (email and password).

- Account verification via email or SMS must be enforced before full access is granted.
- Role-based access (e.g., standard user) must be supported to restrict sensitive functions.
- **Data Security:**
 - All data transmissions between the mobile app, backend, and database must be encrypted using **HTTPS/TLS**.
 - User passwords must be stored securely using hashing and salting.
 - Personal data (profile details, progress, assessments) must be protected against unauthorized access.
- **Privacy:**
 - The app must comply with standard privacy policies (e.g., **GDPR-like principles**).
 - User data must not be shared with third parties without explicit consent.
 - Users must have the ability to delete their accounts and associated data.
- **System Security:**
 - Session management must use secure tokens (e.g., JWT) with expiration to prevent hijacking.
 - Multiple failed login attempts must trigger account lockout or captcha verification.
 - The system must be regularly updated to patch vulnerabilities in third-party libraries and frameworks.
- **Compliance:**

All third-party services must meet standard **security and privacy certifications**

5.4. User Documentation

- A **User Manual** will guide users through system features and functionalities.
- **Tutorial Videos** will be available for complex operations, such as configuring AI weights or writing promotion criteria on the blockchain.
- **Context-Sensitive Help** will display relevant guidance based on the user's current action or page.
- A **Troubleshooting Guide** will help users address common issues.

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7. **Appendices**

PSL website by the Family Educational Services Foundation (FESF) used for dataset and lesson plans:
<https://psl.org.pk/>